A

Boundary vector cells

Encode allocentric boundary vectors



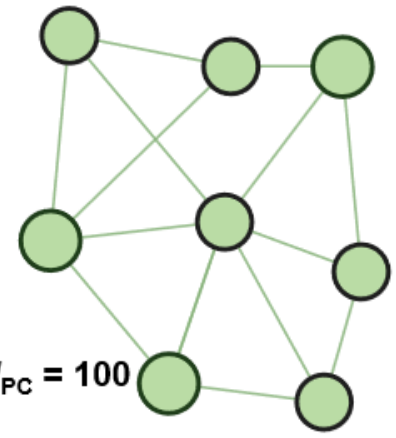
Example tuning profiles

$N_{BVC} = 100$

W^{PB}

Place cells

Multimodal integration



$N_{PC} = 100$

W^{PP} Recurrent excitation

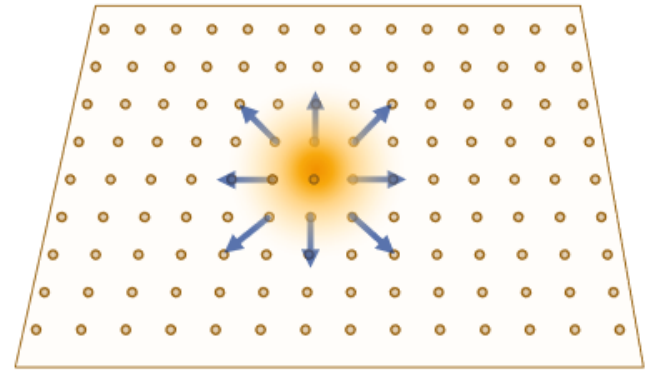
W^{PG}

W^{GP}

Grid cells

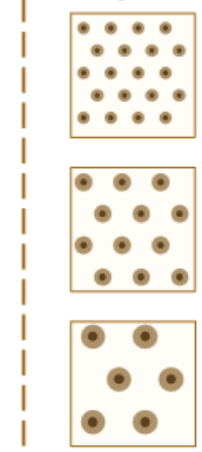
Path integration

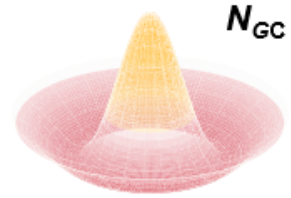
2D neural sheet (neuron lattice)



$N_{GC} = 64 \times 64 \times 3$

Grid pattern







W^{GG} Fixed within-module Mexican-hat connectivity

Global inhibition

$$I_{inh} = \delta \sum_{j=1}^{N_{PC}} s_j^P$$

Summed place-cell activity

Speed-modulated directional drive (Self-motion)

B

Familiar training

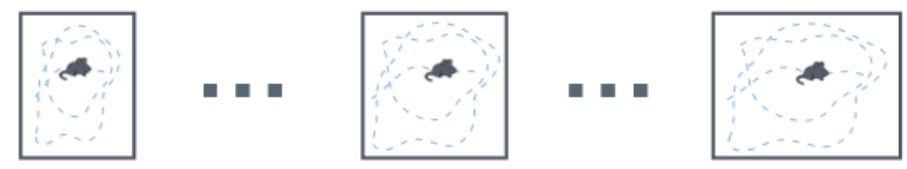
100 × 100 cm → 100 × 100 cm



6 sessions × 1000-s

Initial deformation

70 × 100 cm … 100 × 100 cm … 130 × 100 cm



7 conditions · 10-cm increments · One 1000-s session per condition

● Plastic ○ Frozen

Pathway	Training	Initial exposure	Familiarization
W^{PB}	●	○	●
W^{PG}	●	○	●
W^{GP}	●	○	●
W^{PP}	●	○	○
W^{GG}	Fixed connectivity throughout		

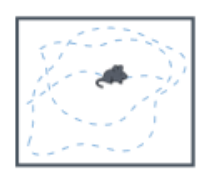
Deformation familiarization

80 × 100 cm



Compressed

120 × 100 cm



Elongated

Sessions 0 Familiarization → Sessions 1-9 Familiarization